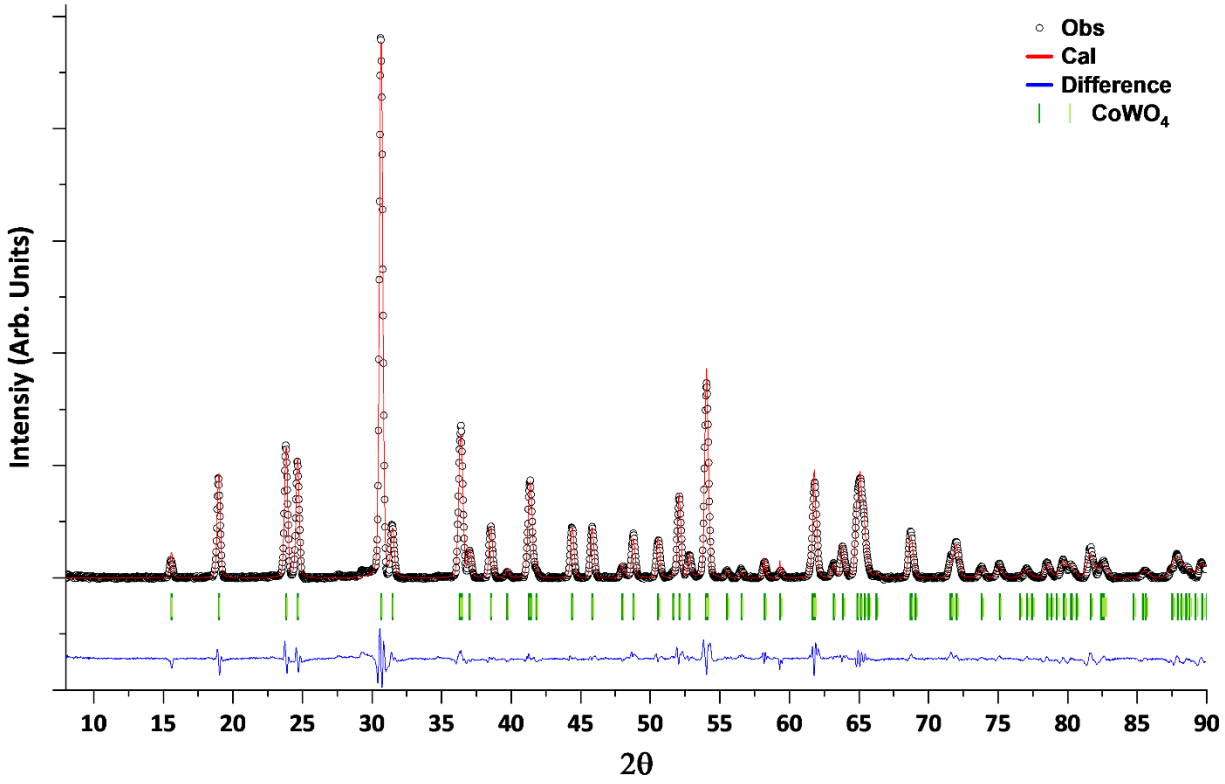
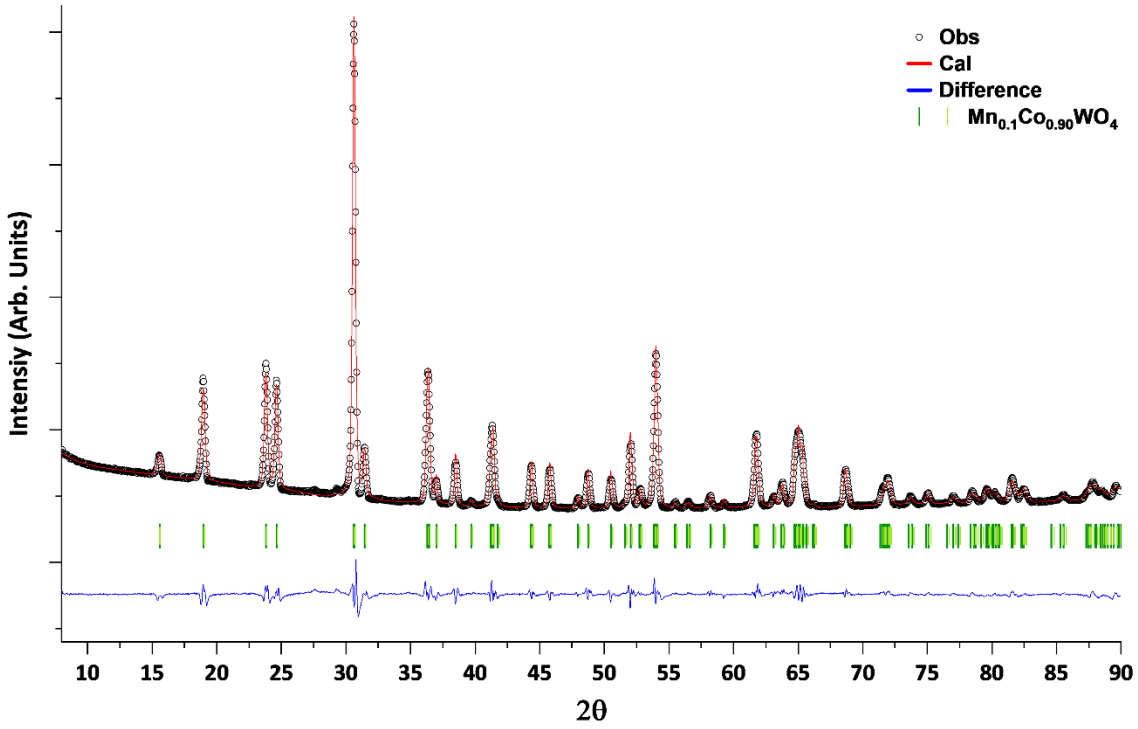
Mn Incorporation and Mechanical Activation Synergistically Boost the Oxygen Evolution Reaction (OER) Activity of Cobalt Tungstate

Andrea Griesi[1], Manjunath Chatti[2], Luca Saluta[2], Michele Ferri[2], Lea Pasquale[3], Dorwal Marchelli[5], Ambra Celotto[1,4], Francesco De Boni[3], Andrea Sala[6], Alexandre G. Brolo[7], Roman Krahne[5], Mauro Gemmi[6], Liberato Manna[2], Giorgio Divitini[1]**

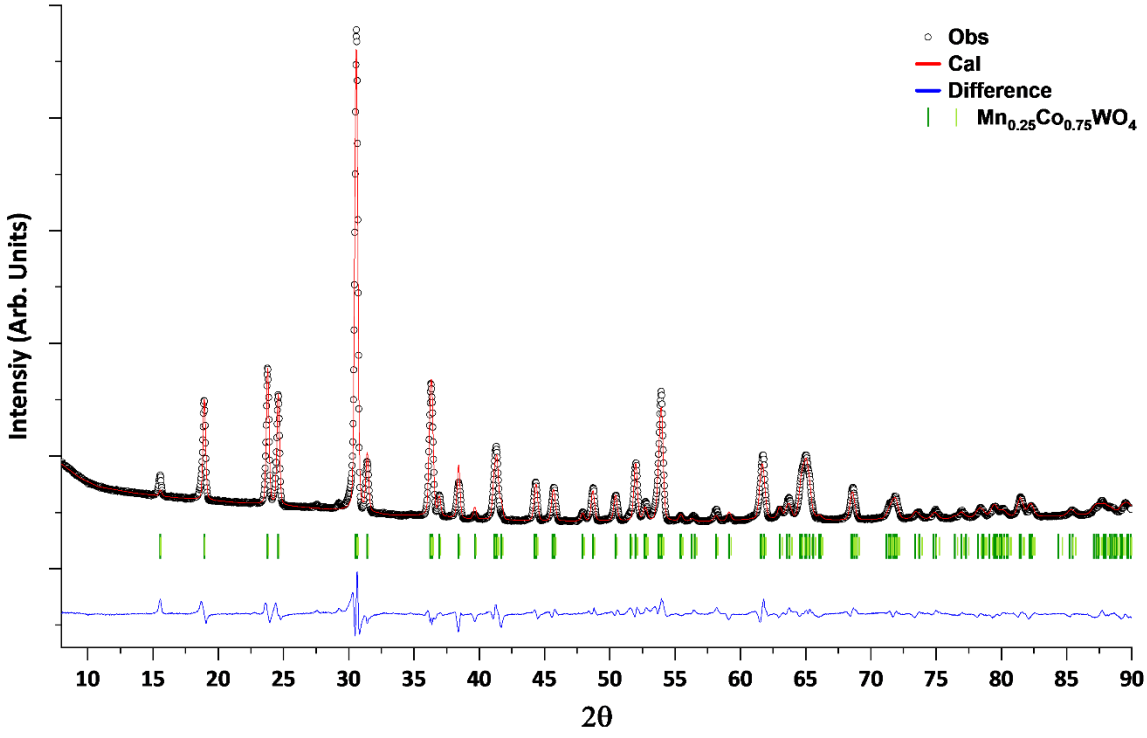
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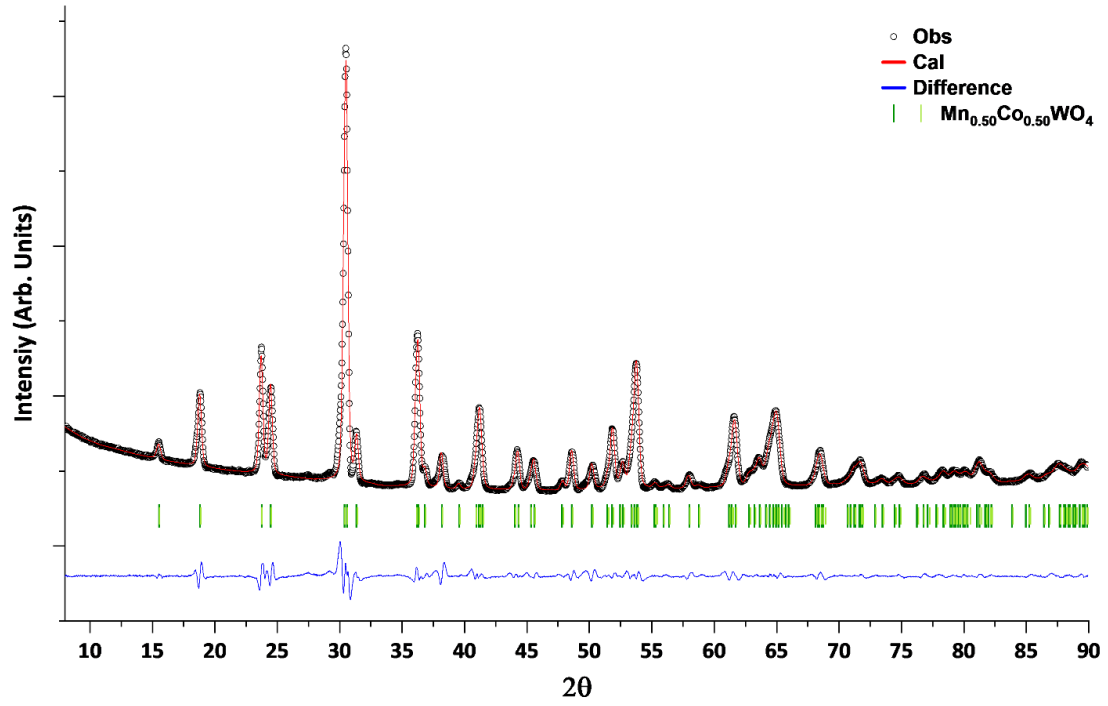
S1: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=0%, on the X-ray diffraction data.



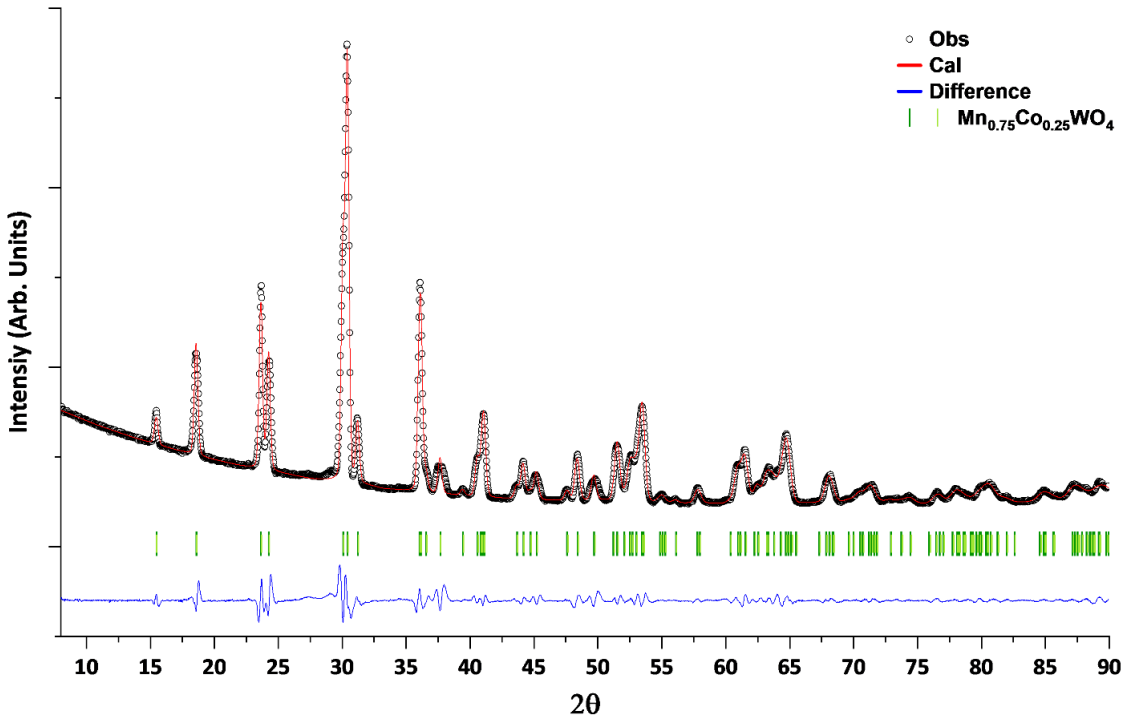
S2: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=10%, on the X-ray diffraction data.



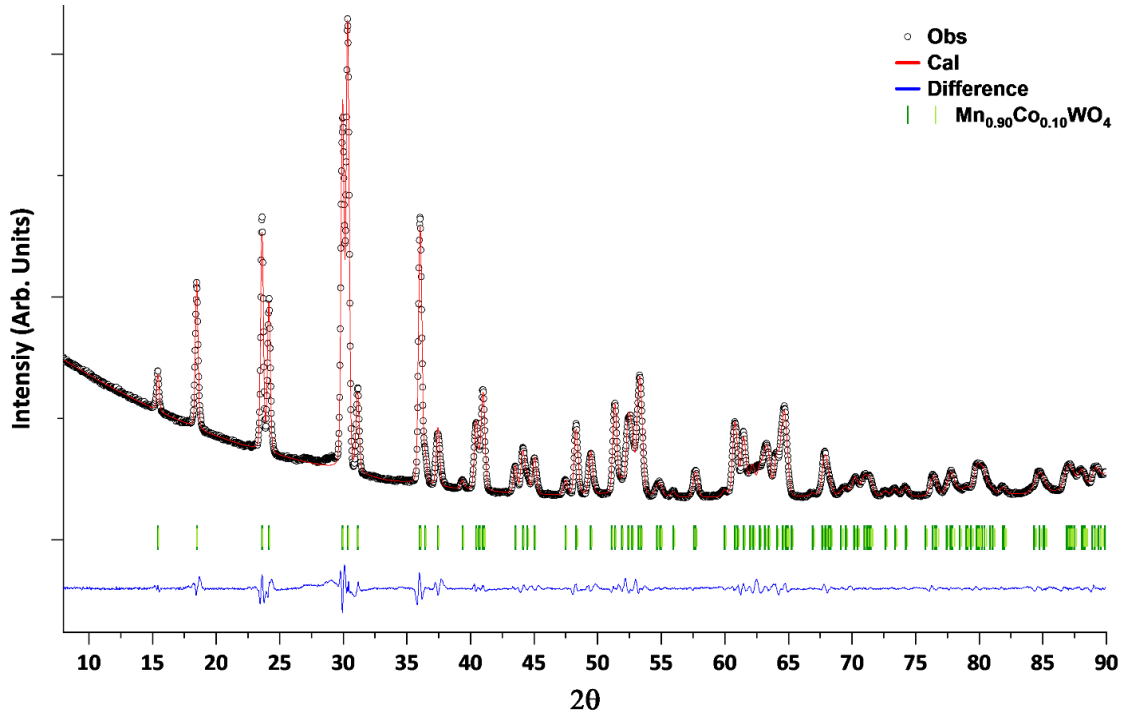
S3: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=25%, on the X-ray diffraction data.



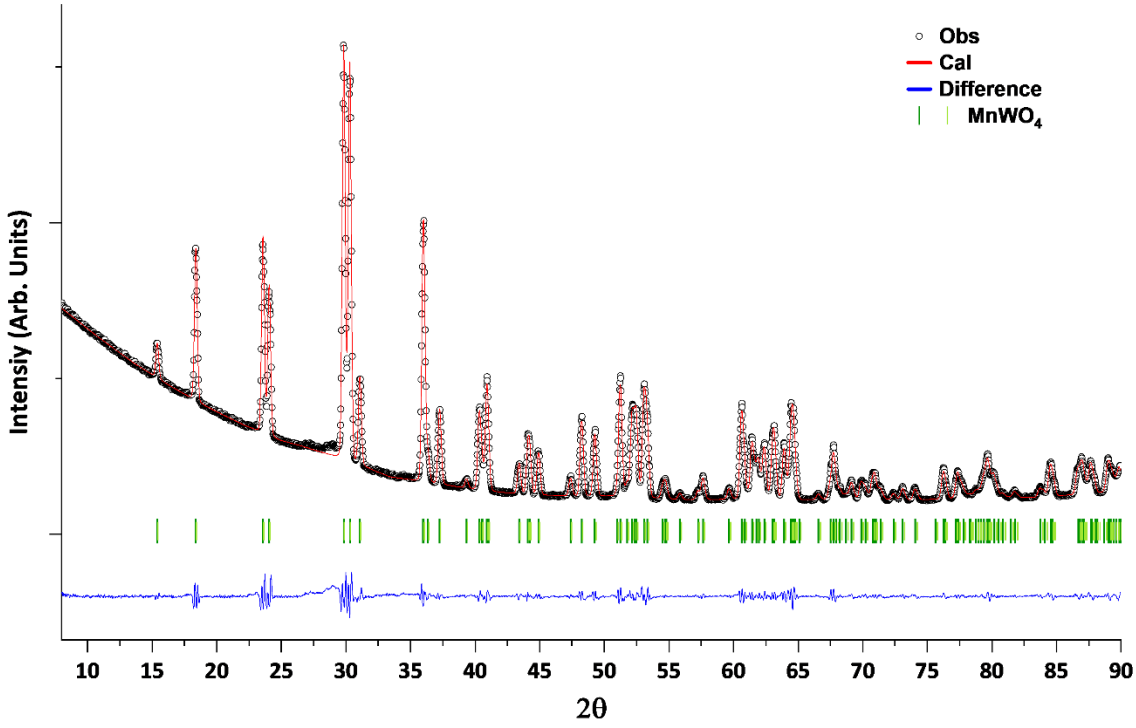
S4: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=50%, on the X-ray diffraction data.



S5: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=75%, on the X-ray diffraction data.



S6: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=90%, on the X-ray diffraction data.



S7: Rietveld refinement of $Mn_xCo_{(1-x)}WO_4$ with x=100%, on the X-ray diffraction data.

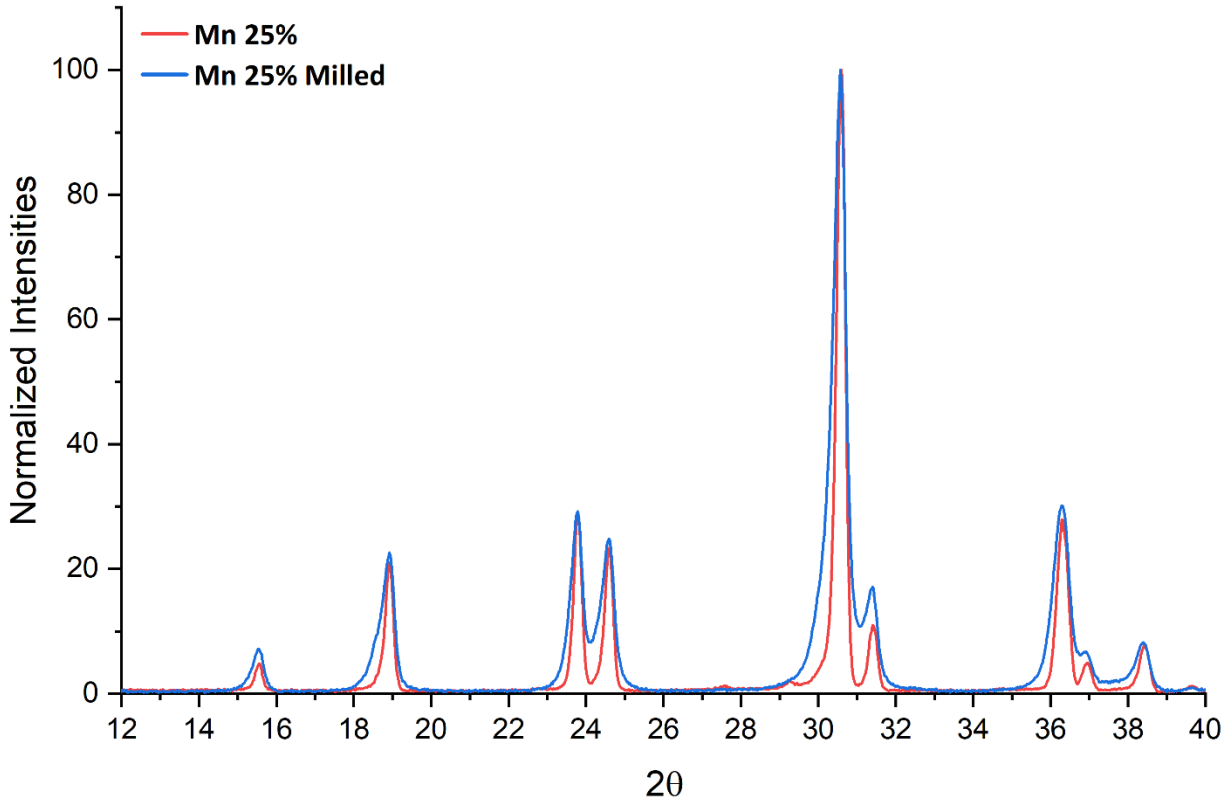
Table S1: cell parameters after Rietveld refinement.

	a (Å)	b(Å)	c(Å)	β (°)	Vol (Å^3)
Mn_0%	4.667(9)	5.679(9)	4.945(6)	90.0(1)	131.12
Mn_10%	4.674(3)	5.685(1)	4.948(4)	90.2(2)	131.49
Mn_25%	4.684(1)	5.690(1)	4.953(1)	90.2(7)	132.01
Mn_50%	4.709(9)	5.703(3)	4.962(7)	90.4(7)	133.30
Mn_75%	4.769(8)	5.726(7)	4.980(1)	90.8(5)	136.01
Mn_90%	4.801(2)	5.740(2)	4.987(3)	91.0(1)	137.41
Mn_100%	4.825(6)	5.751(6)	4.991(7)	91.1(8)	138.51

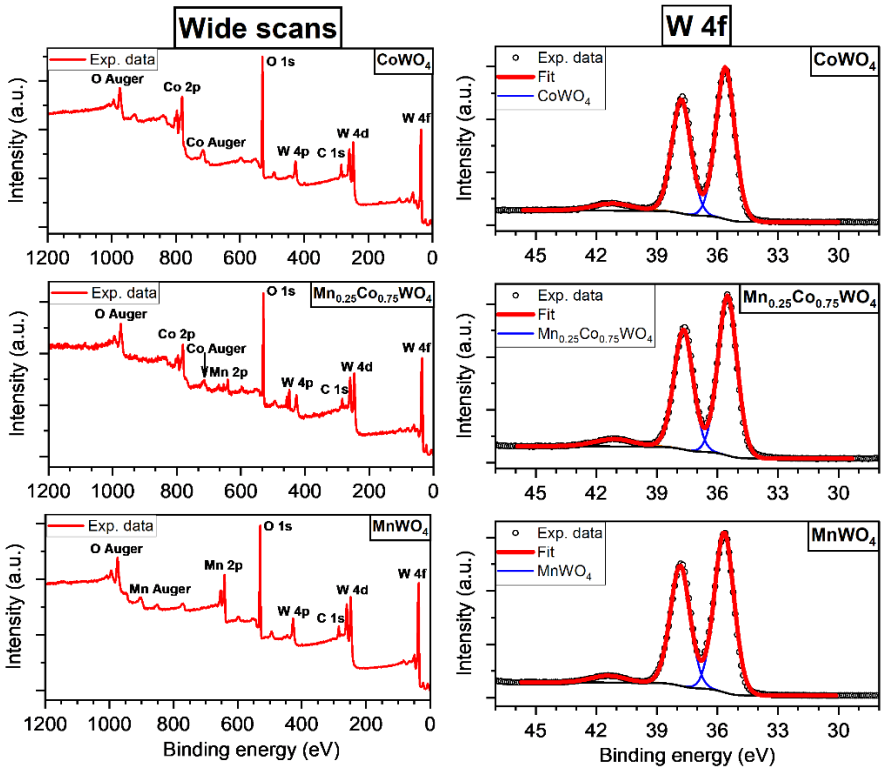
Table S2: The overall reliability factors of Rietveld refinement

	R_p*	wR_p*	GOF*
Mn_0%	0.018	0.027	5.74
Mn_10%	0.019	0.030	5.16
Mn_25%	0.021	0.032	5.48
Mn_50%	0.019	0.034	5.63
Mn_75%	0.025	0.041	5.94
Mn_90%	0.019	0.028	3.71
Mn_100%	0.019	0.029	3.40

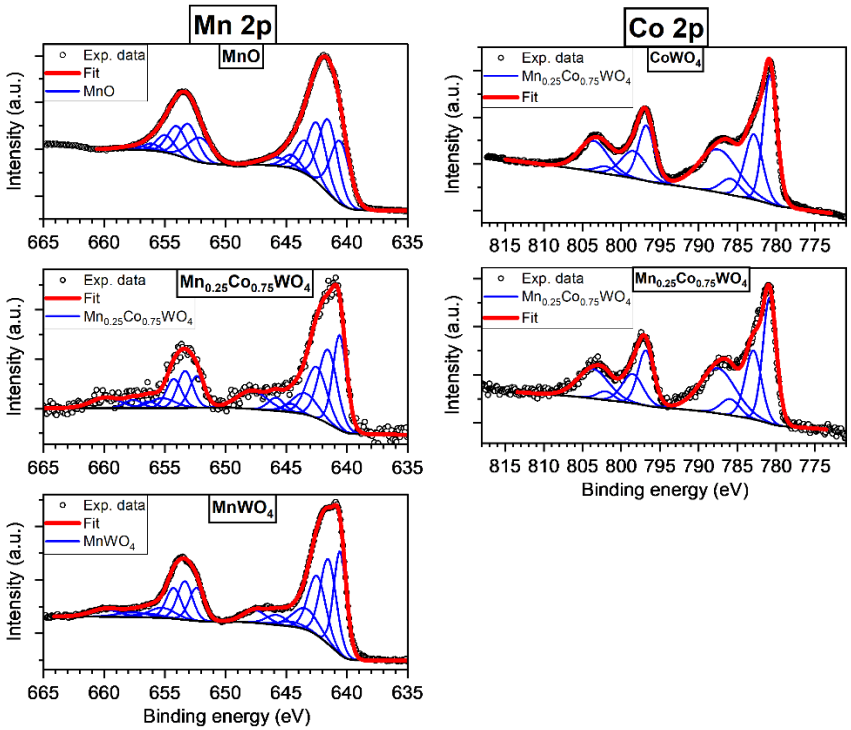
** Rp, wRp, and GOF are the profile, weighted profile, and the goodness of fit.*



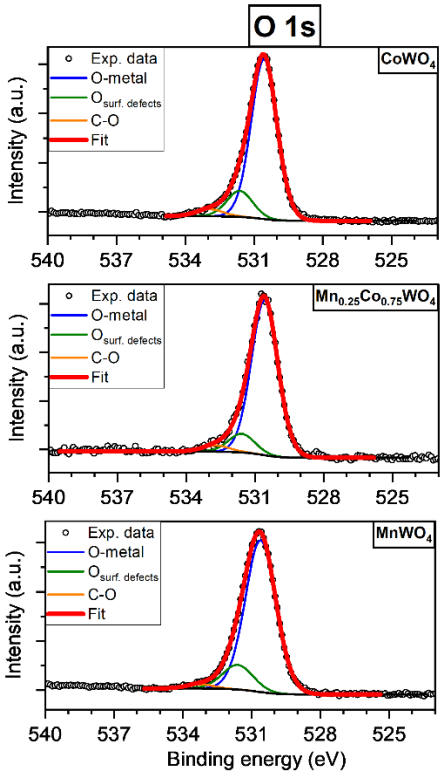
S8: Diffraction patterns with normalised intensities for $Mn_{0.25}Co_{0.75}WO_4$ samples before (red) and after (blue) ball milling.



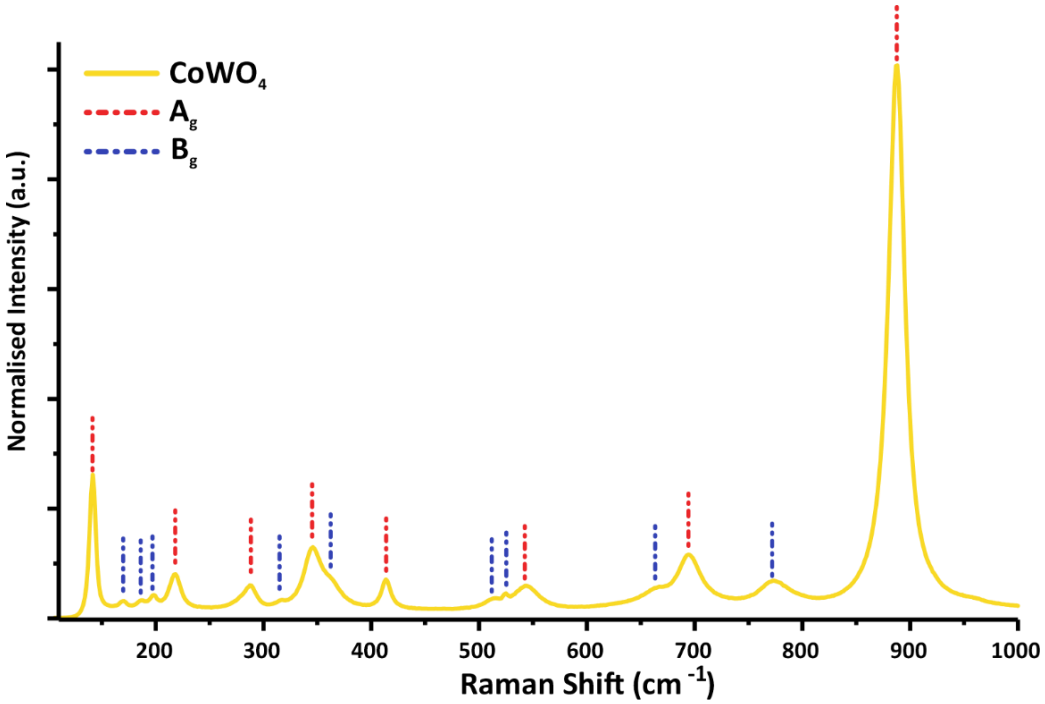
S9: Full XPS spectra of $CoWO_4$, $Mn_{0.25}Co_{0.75}WO_4$, $MnWO_4$ (left) and detail of the tungsten 4f feature (right).



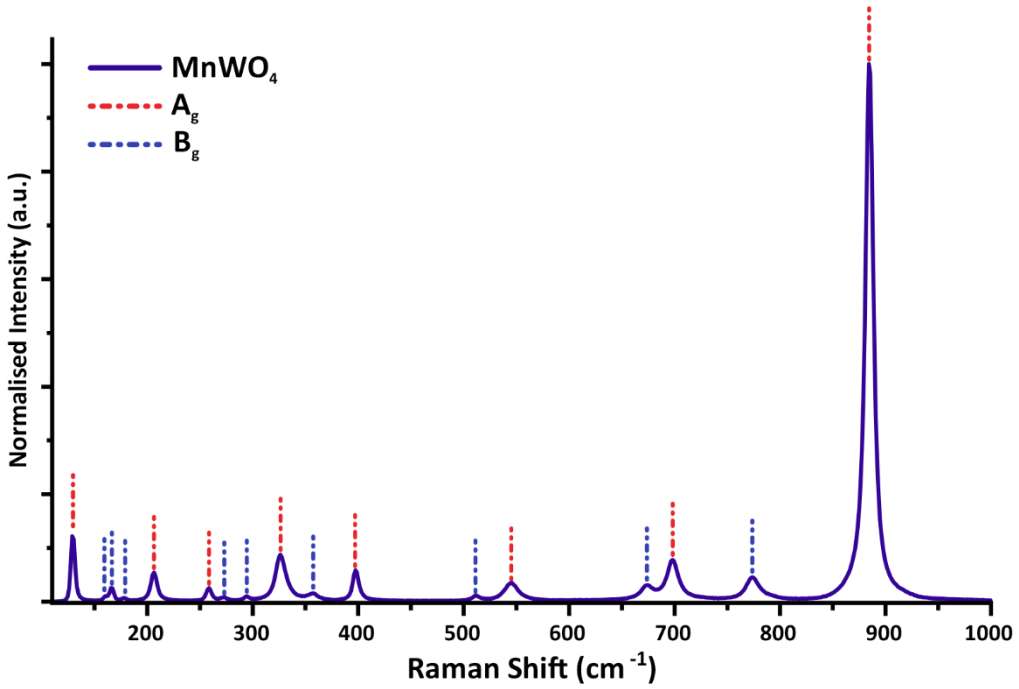
S10: Magnified view of the XPS signal of manganese in MnO, $Mn_{0.25}Co_{0.75}WO_4$, $MnWO_4$ (left) and cobalt in $CoWO_4$ and $Mn_{0.25}Co_{0.75}WO_4$ (right).



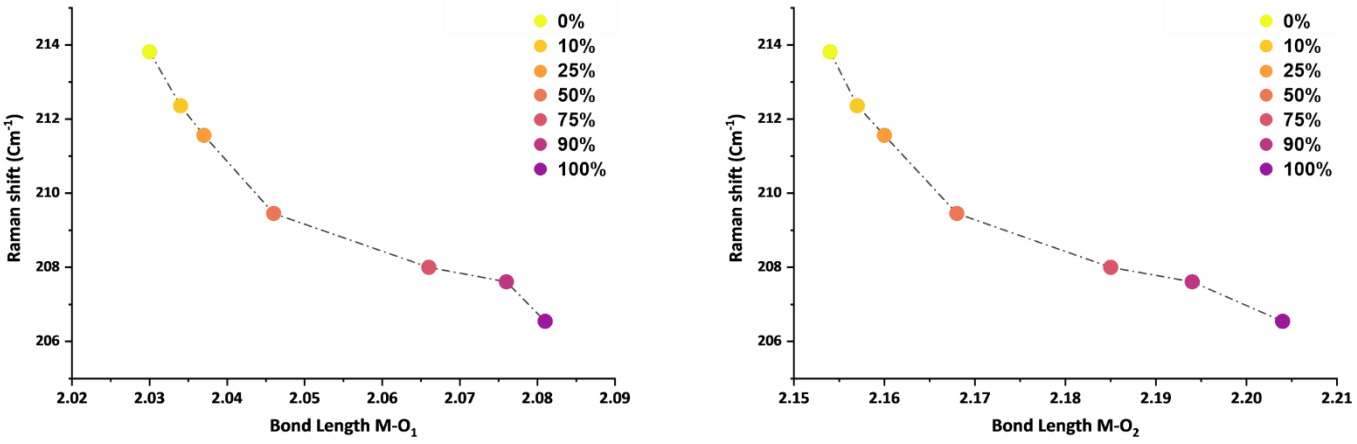
S11: Magnified view of XPS spectra of oxygen in $CoWO_4$, $Mn_{0.25}Co_{0.75}WO_4$, $MnWO_4$.



S12: Raman spectrum of pristine $CoWO_4$ with A_g modes highlighted in red and B_g modes in blue[1]. Seventeen Raman-active modes are identified, consistent with group-theoretical predictions; one mode is not observed as it lies beyond the measured spectral window.



S13: Raman spectrum of pristine $MnWO_4$ with A_g modes highlighted in red and B_g modes in blue[2]. Seventeen Raman-active modes are identified, consistent with group-theoretical predictions; one mode is not observed as it lies beyond the measured spectral window.



S14:

Table S3: (a-g) Elemental concentrations (normalised at%) obtained by averaging five measurement positions (P1-P5) for each sample with different Mn content, before and after ball milling, measured by μ-XRF. (h)μ-XRF-derived Mn/(Mn + Co) cationic ratio after ball milling.

a)

Before Ball Milling			
Mn 0%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	n.d.*	52.1	47.9
P2	n.d.	52.1	47.9
P3	n.d.	52.1	47.9
P4	n.d.	52.1	47.9
P5	n.d.	52.1	47.9
Mean value	-	**52.1**	**47.9**
Std dev.***	-	**0.0**	**0.0**
Std dev. Rel (%)	-	**0.0**	**0.0**
Conf. interval.	-	**0.0**	**0.0**

After Ball Milling			
Mn 0%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	n.d.	52.6	47.4
P2	n.d.	52.6	47.4
P3	n.d.	52.4	47.6
P4	n.d.	52.6	47.4
P5	n.d.	52.6	47.4
Mean value	-	**52.6**	**47.4**
Std dev.***	-	0.1	0.1
Std dev. Rel (%)	-	0.1	0.1
Conf. interval.	-	0.0	0.0

b)

Before Ball Milling			
Mn 10%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	4.7	46.7	48.7
P2	11.9	40.1	47.9
P3	1.6	51.3	47.2
P4	70.4	11.9	17.7
P5	1.8	50.1	48.1
Mean value	**18.1**	**40.0**	**41.9**
Std dev.***	29.5	16.3	13.5
Std dev. Rel (%)	163.5	40.8	32.3
Conf. interval.	13.2	7.3	6.0

After Ball Milling			
Mn 10%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	5.4	45.8	48.8
P2	5.7	45.6	48.7
P3	5.5	45.7	48.9
P4	5.7	45.6	48.7
P5	5.4	45.7	48.9
Mean value	**5.5**	**45.6**	**48.8**
Std dev.***	0.2	0.1	0.1
Std dev. Rel (%)	2.8	0.3	0.2
Conf. interval.	0.1	0.1	0.0

c)

Before Ball Milling			
Mn 25%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	6.7	45.2	48.1
P2	5.2	46.0	48.8
P3	13.4	36.2	50.4
P4	4.4	47.5	48.2
P5	5.9	45.6	48.5
Mean value	***7.1***	***44.1***	***48.8***
Std dev.***	*3.6*	*4.5*	*0.9*
Std dev. Rel (%)	*50.7*	*10.2*	*1.9*
Conf. interval.	*1.6*	*2.0*	*0.4*

After Ball Milling			
Mn 25%	**Mn (at%)**	**Co (at%)**	**W (at%)**
P1	12.7	40.1	47.2
P2	12.4	40.3	47.3
P3	13.0	40.0	47.0
P4	12.5	40.3	47.2
P5	12.6	40.2	47.2
Mean value	***12.6***	***40.2***	***47.2***
Std dev.***	*0.2*	*0.1*	*0.1*
Std dev. Rel (%)	*1.7*	*0.3*	*0.2*
Conf. interval.	*0.1*	*0.1*	*0.0*

d)

Before Ball Milling

Mn 50%	Mn (at%)	Co (at%)	W (at%)
P1	19.6	31.8	48.6
P2	29.6	19.6	50.9
P3	17.3	33.9	48.8
P4	20.4	28.3	51.3
P5	17.8	32.6	49.6
Mean value	***20.9***	***29.2***	***49.8***
*Std dev.****	5.0	5.8	1.2
Std dev. Rel (%)	23.9	19.8	2.4
Conf. interval.	2.2	2.6	0.5

After Ball Milling

Mn 50%	Mn (at%)	Co (at%)	W (at%)
P1	24.9	28.3	46.8
P2	26.0	27.2	46.8
P3	25.0	27.8	47.1
P4	22.4	30.0	47.6
P5	25.4	27.8	46.8
Mean value	***24.8***	***28.2***	***47.0***
*Std dev.****	1.4	1.1	0.3
Std dev. Rel (%)	5.6	3.8	0.7
Conf. interval.	0.6	0.5	0.2

e)

Before Ball Milling

Mn 75%	Mn (at%)	Co (at%)	W (at%)
P1	37.5	13.1	49.4
P2	32.2	18.6	49.2
P3	32.6	18.1	49.3
P4	30.2	20.8	49.0
P5	32.4	18.3	49.3
Mean value	***33.0***	***17.8***	***49.2***
*Std dev.****	2.7	2.8	0.1
Std dev. Rel (%)	8.2	15.9	0.3
Conf. interval.	1.2	1.3	0.1

After Ball Milling

Mn 75%	Mn (at%)	Co (at%)	W (at%)
P1	37.7	13.2	49.1
P2	37.3	13.7	49.0
P3	37.7	13.1	49.1
P4	37.5	13.5	49.1
P5	37.4	13.5	49.1
Mean value	***37.5***	***13.4***	***49.1***
*Std dev.****	0.2	0.2	0.1
Std dev. Rel (%)	0.5	1.7	0.1
Conf. interval.	0.1	0.1	0.0

f)

Before Ball Milling

Mn 90%	Mn (at%)	Co (at%)	W (at%)
P1	41.7	8.6	49.7
P2	43.9	6.6	49.5
P3	40.9	9.7	49.4
P4	41.7	8.6	49.7
P5	42.2	8.2	49.6
Mean value	***42.1***	***8.4***	***49.5***
*Std dev.****	1.1	1.1	0.1
Std dev. Rel (%)	2.7	13.5	0.2
Conf. interval.	0.5	0.5	0.0

After Ball Milling

Mn 90%	Mn (at%)	Co (at%)	W (at%)
P1	47.2	5.8	47.1
P2	47.5	5.7	46.8
P3	47.5	5.7	46.8
P4	47.1	5.8	47.1
P5	47.1	5.8	47.0
Mean value	***47.3***	***5.8***	***47.0***
*Std dev.****	0.2	0.1	0.1
Std dev. Rel (%)	0.5	1.3	0.3
Conf. interval.	0.1	0.0	0.1

g)

Before Ball Milling

Mn 100%	Mn (at%)	Co (at%)	W (at%)
P1	49.1	n.d.	50.9
P2	50.6	n.d.	49.4
P3	50.6	n.d.	49.4
P4	50.3	n.d.	49.7
P5	50.3	n.d.	49.7
Mean value	**50.2**	**-**	**49.8**
Std dev.***	0.6	-	0.6
Std dev. Rel (%)	1.3	-	1.3
Conf. interval.	0.3	-	0.3

After Ball Milling

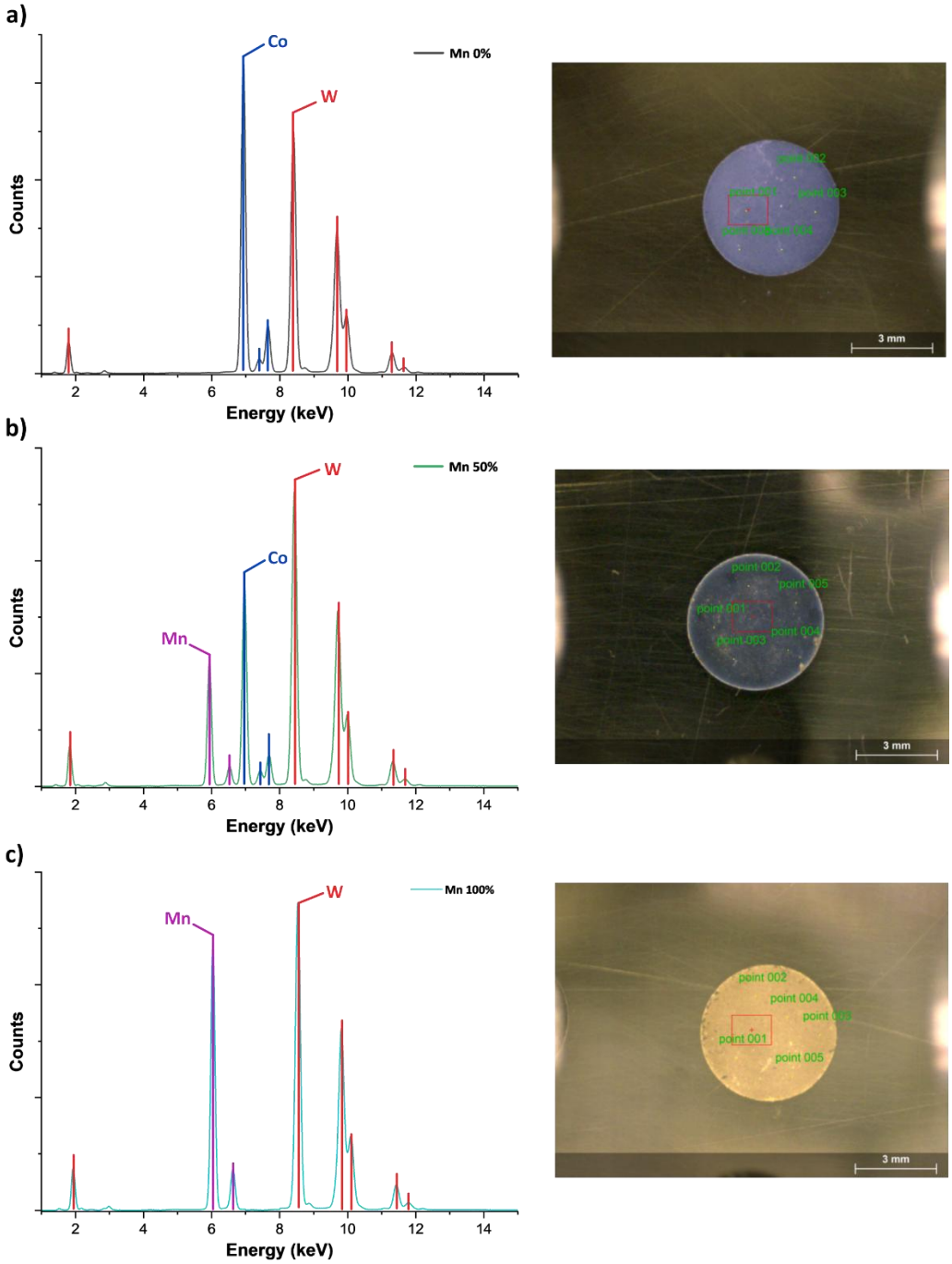
Mn 100%	Mn (at%)	Co (at%)	W (at%)
P1	52.0	n.d.	48.0
P2	51.7	n.d.	48.3
P3	51.8	n.d.	48.2
P4	52.1	n.d.	47.9
P5	51.7	n.d.	48.3
Mean value	**51.8**	**-**	**48.2**
Std dev.***	0.2	-	0.2
Std dev. Rel (%)	0.3	-	0.4
Conf. interval.	0.1	-	0.1

h)

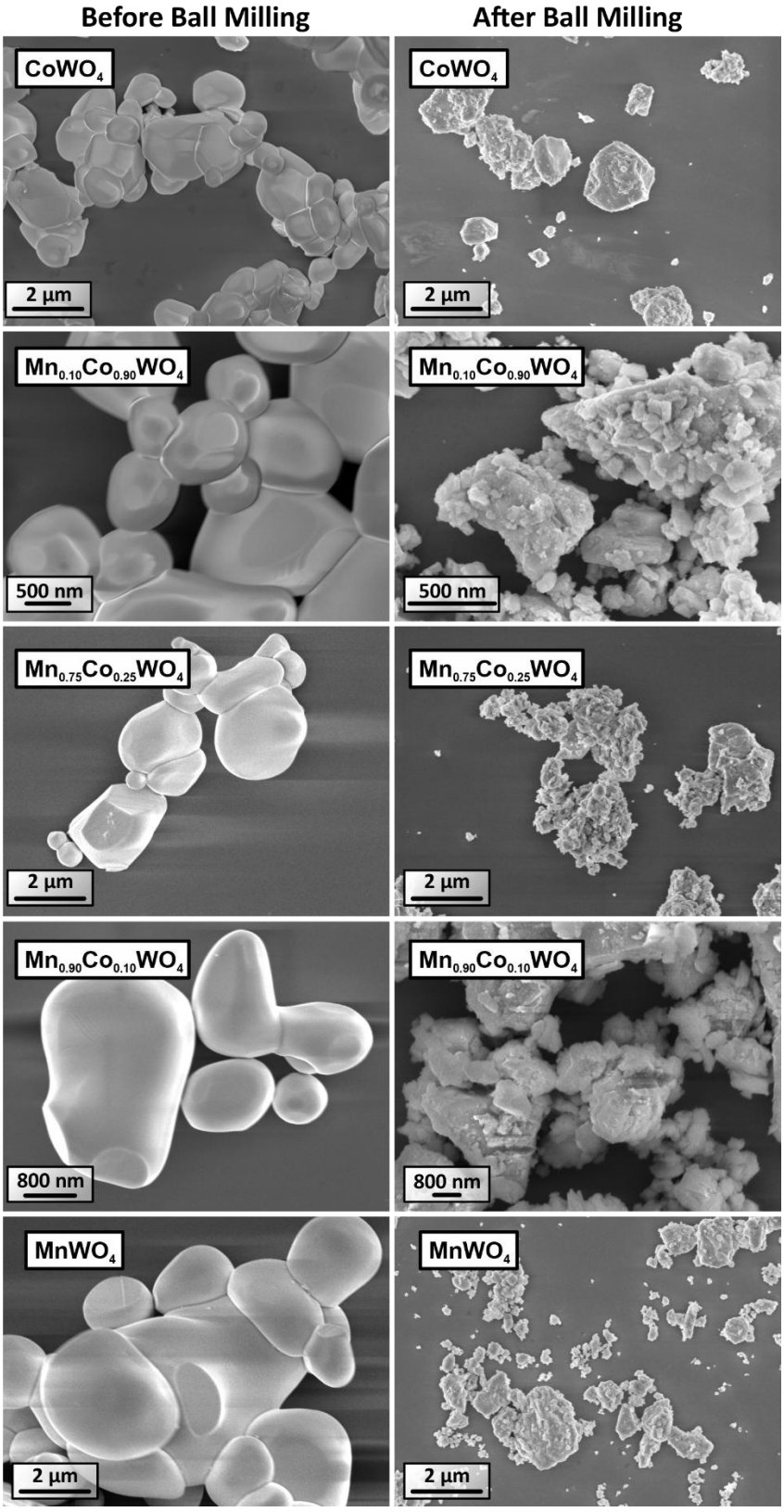
After Ball Milling

Nominal Mn content	Mn/(Mn+Co)
0%	0.00
10%	0.11
25%	0.24
50%	0.47
75%	0.74
90%	0.89
100%	1.00

** n.d. = not detected (below the μ-XRF detection limit).*
***Atomic concentrations are normalised to the elements detected by μ-XRF (oxygen is not detectable with the employed setup).*
**** Std Dev. represents the standard deviation across five measurement points; the overall analytical uncertainty of μ-XRF atomic concentrations is estimated at ~5–10%.*



S14: Representative μ-XRF spectra of the (a) $Co_{1.05}W_{0.95}O_4$, (b) $Mn_{0.50}Co_{0.56}W_{0.94}O_4$, and (c) $Mn_{1.04}W_{0.96}O_4$ pellets, highlighting the W-M_α (1.78 keV), Mn-K_α (5.90 keV), Co-K_α (6.93 keV), and W-L_α (8.40 keV) emission lines. The corresponding optical images of the pellets, with the five analysed points marked, are shown.



S15: SEM images of samples $CoWO_4$, $Mn_{0.10}Co_{0.90}WO_4$, $Mn_{0.50}Co_{0.50}WO_4$, $Mn_{0.90}Co_{0.10}WO_4$, $MnWO_4$, before (left) and after (right) ball milling.

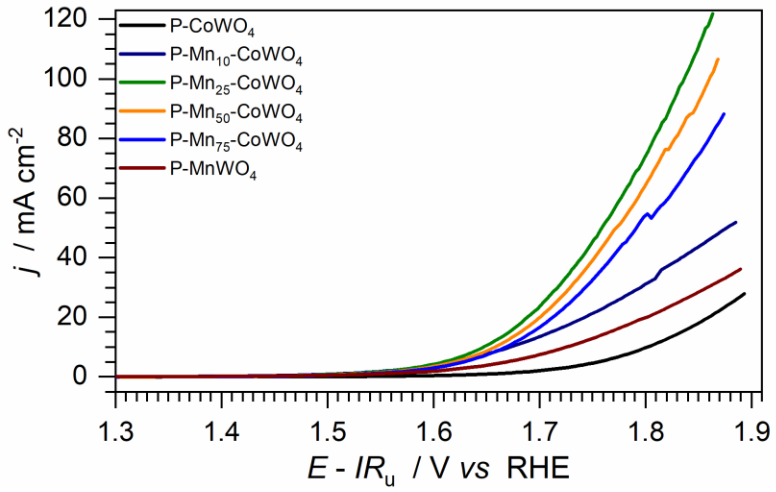


Figure S16: Electrochemical OER performance of $Mn_xCo_{1-x}WO_4$ catalysts before ball milling. Quasi-stabilised linear sweep voltammograms (scan rate = 0.005 V s^{-1}; 5th scans shown) recorded in a three-electrode configuration in 0.5 M H_2SO_4 at 23 ± 2 °C. Data were manually post-corrected for the iR_u drop using uncompensated resistance values derived from electrochemical impedance spectroscopy. Currents are normalised to the geometric surface area of the electrodes.

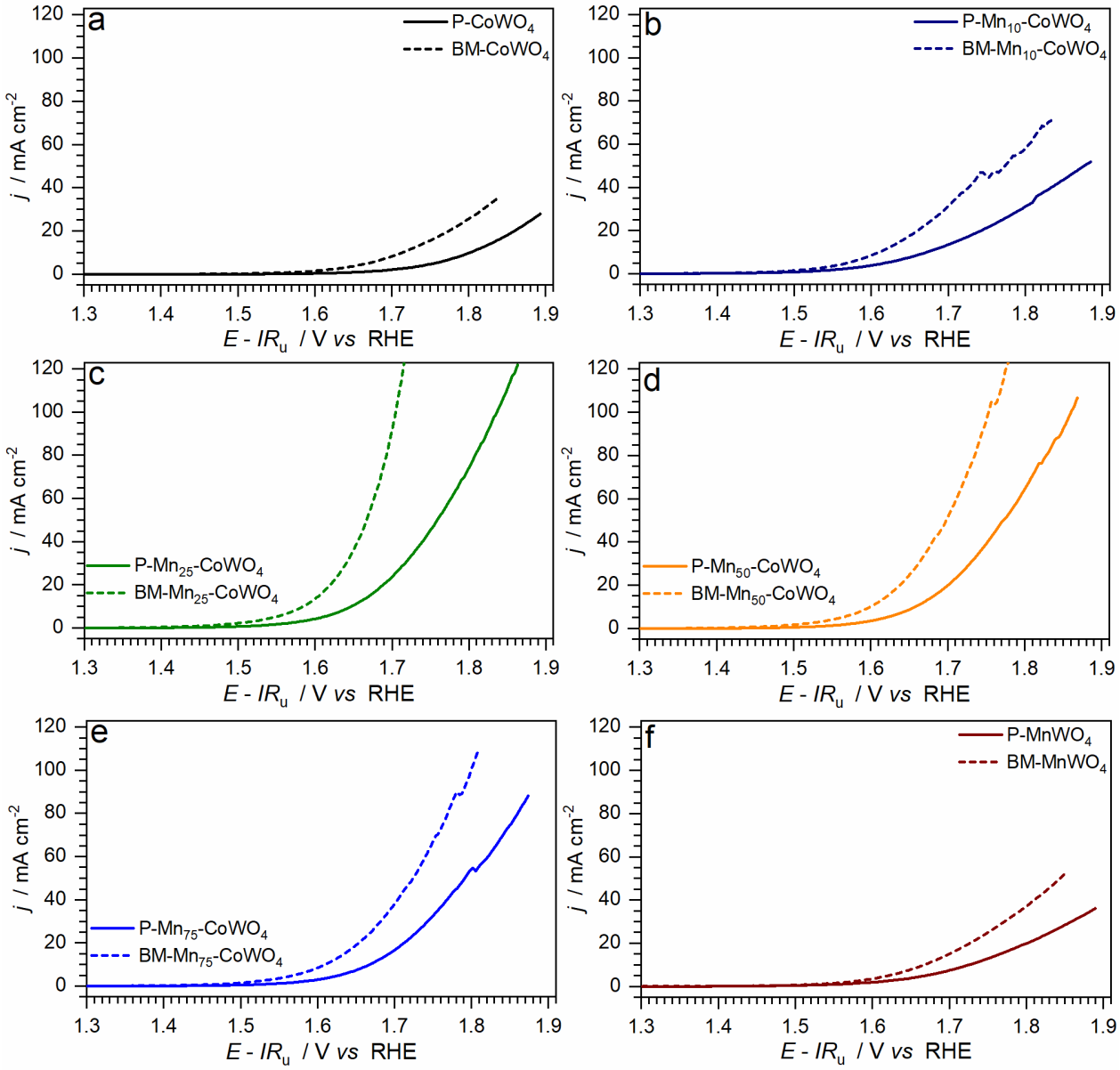


Figure S17:Electrochemical OER performance of ball-milled $Mn_xCo_{1-x}WO_4$ catalysts. Comparison of quasi-stabilized linear sweep voltammograms (scan rate = 0.005 V s^{-1}; 5th scans shown) for ball-milled $Mn_xCo_{1-x}WO_4$ catalysts with Mn contents of (a) x = 0, (b) x = 0.10, (c) x = 0.25, (d) x = 0.50, (e) x = 0.75, and (f) x = 1.00. OER electrocatalytic activity measured in a three-electrode configuration in 0.5 M H_2SO_4 at 23 ± 2 °C: For comparison, the corresponding pre-ball-milled catalysts are also included in each panel. Data were manually post-corrected for the iR_u drop using uncompensated resistance values derived from electrochemical impedance spectroscopy. Currents are normalised to the geometric surface area of the electrodes.

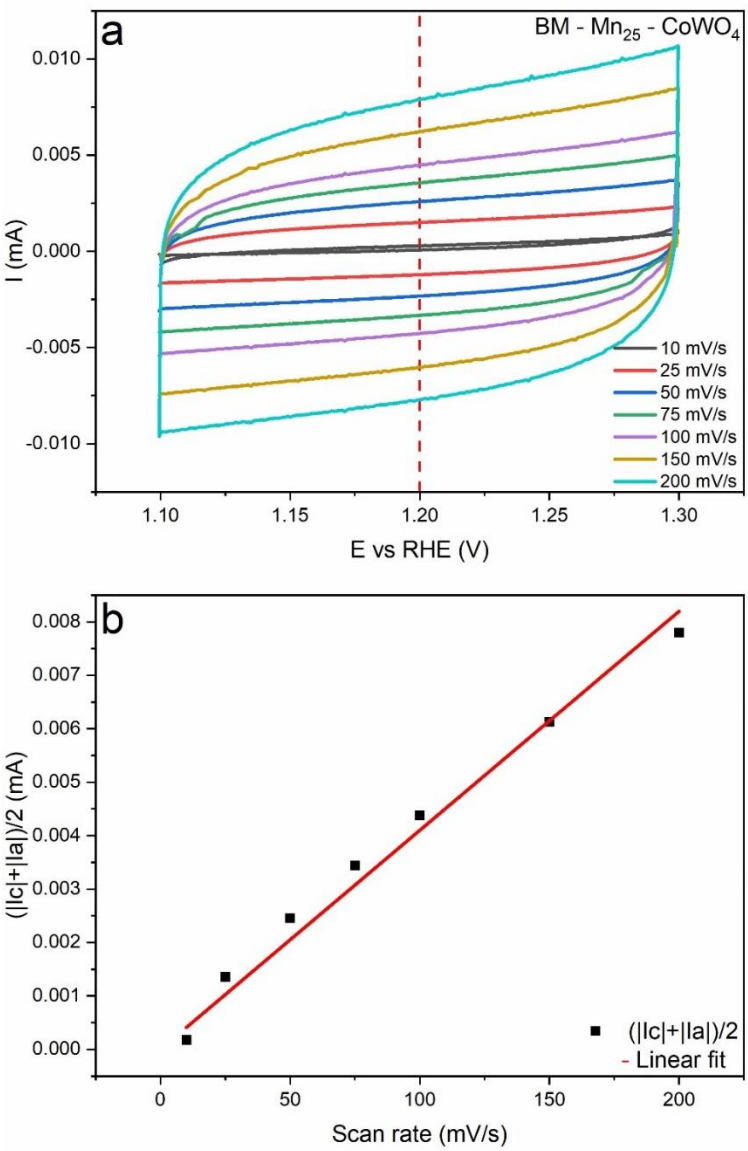


Figure S18: Determination of the double-layer capacitance (Cdl) of the ball-milled $Mn_{0.25}Co_{0.75}WO_4$ catalyst. (a) Cyclic voltammograms recorded in the non-faradaic potential region (1.10–1.31 V vs RHE) at scan rates of 10, 25, 50, 75, 100, 150, and 200 mV s^{-1} in 0.5 M H_2SO_4 using a three-electrode configuration at 23 ± 2 °C. The dashed red line indicates the potential (1.20 V vs RHE) at which the capacitive currents were extracted for Cdl determination. (b) Plot of the average capacitive current, (|Ic| + |Ia|)/2, as a function of scan rate. The double-layer capacitance (Cdl) was determined from the slope of the linear fit. Measurements were performed using catalyst films deposited on a glassy carbon electrode (geometric area = 0.07 cm^2), and currents are normalised to the geometric surface area of the electrode.

Table S4: Linear-fit double-layer capacitance (Cdl) values of the ball-milled $Mn_xCo_{1-x}WO_4$ catalysts and the corresponding normalisation factors relative to BM-$CoWO_4$.

Sample	Cdl (µF)	Normalisation factor (Cdl/20.6)
BM-Mn0	20.6	1.00
BM-Mn10	33.7	1.64
BM-Mn25	41.0	1.99
BM-Mn50	27.6	1.34
BM-Mn75	27.7	1.34
BM-Mn90	24.6	1.19
BM-Mn100	43.0	2.09

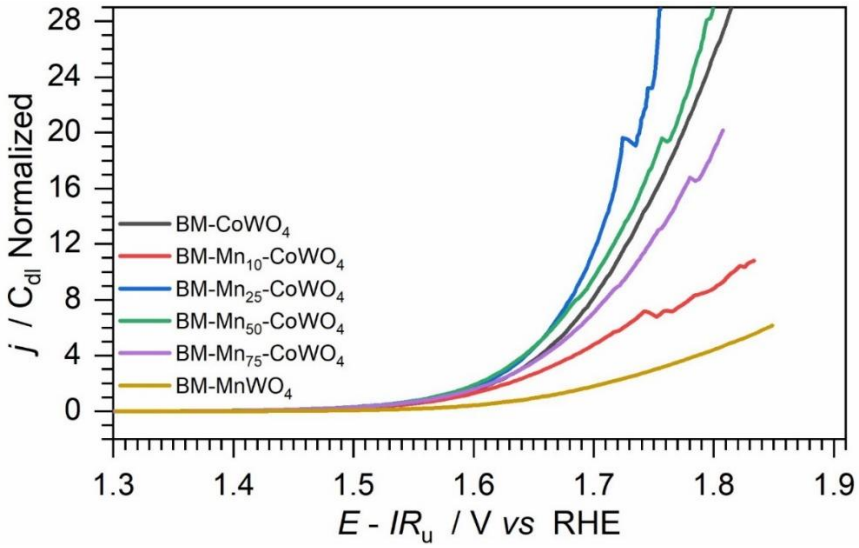


Figure S19: Linear sweep voltammograms (LSVs) of the ball-milled $Mn_xCo_{1-x}WO_4$ catalyst series in 0.5 M H_2SO_4 at 23 ± 2 °C normalised using the C_{dl}-derived factors listed in Table S4. The normalisation factors were obtained from the linear-fit double-layer capacitance values using BM-$CoWO_4$ as the reference sample. Potentials were corrected for ohmic losses using the experimentally determined uncompensated resistance (Ru).

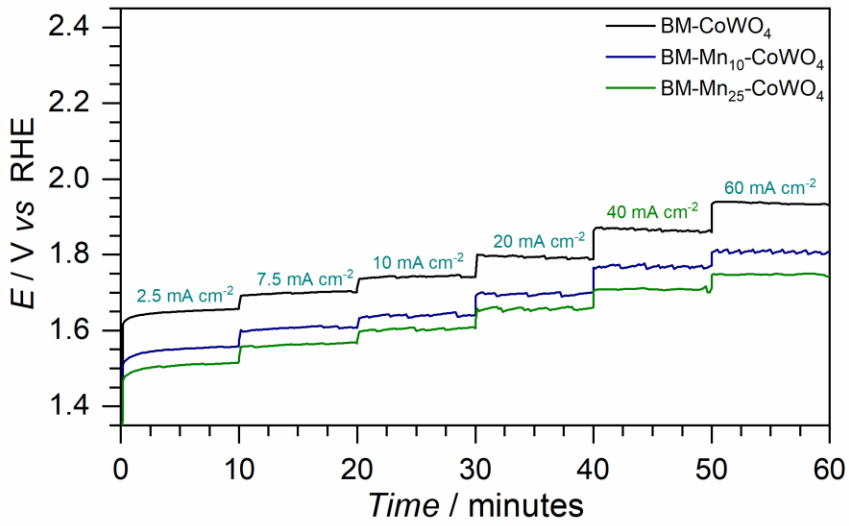


Figure S20: Electrochemical OER activity of selected ball-milled $Mn_xCo_{1-x}WO_4$ catalysts under increasing current density. Stepwise chronopotentiometric measurements of BM-$CoWO_4$, BM-$Mn_{0.1}Co_{0.9}WO_4$, and BM-$Mn_{0.25}Co_{0.75}WO_4$ recorded in a three-electrode configuration in 0.5 M H_2SO_4 at 23 ± 2 °C. The applied current density was sequentially increased from 2.5 to 60 mA cm^{-2}, as indicated in the figure. The currents are normalised to the geometric surface area of the electrodes.

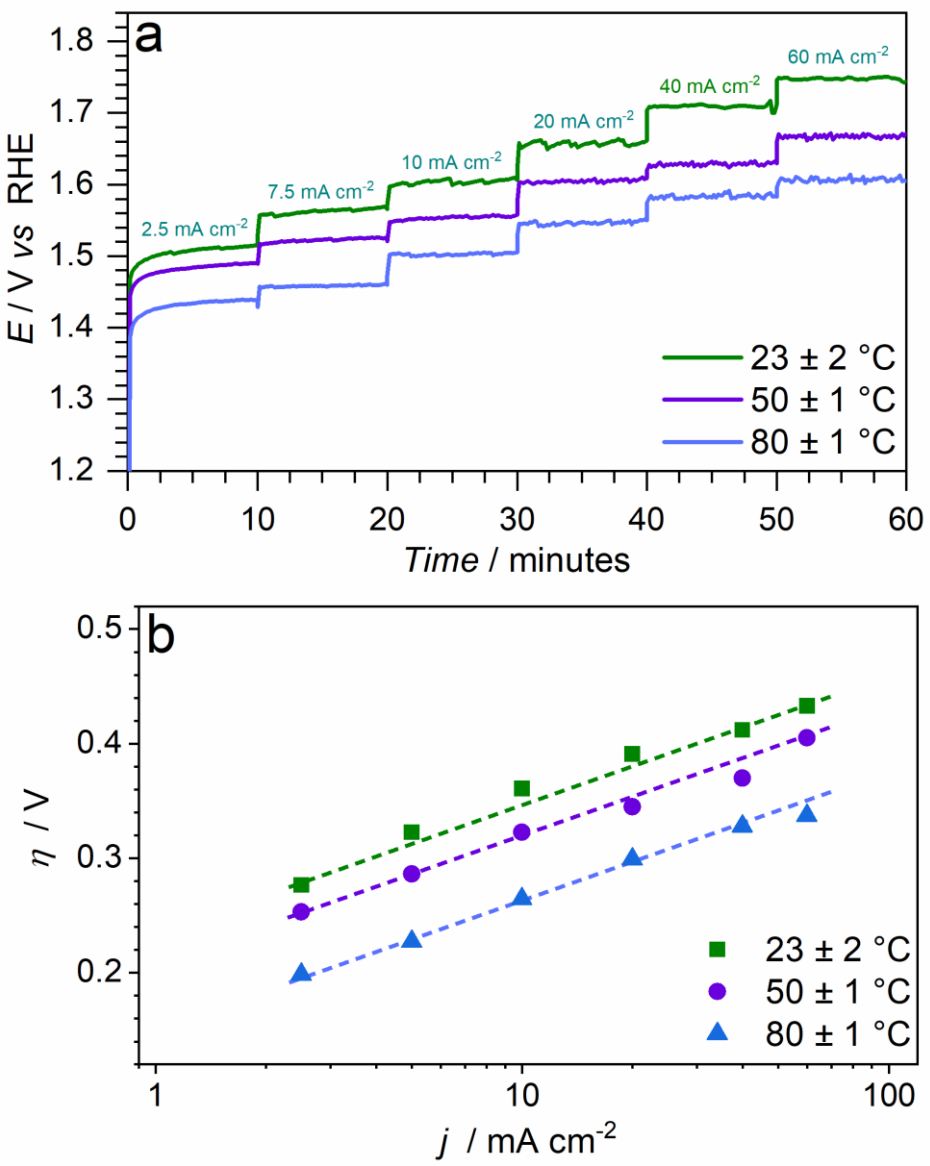


Figure S21: Temperature-dependent OER performance of BM-$Mn_{0.25}Co_{0.75}WO_4$. (a) Stepwise chronopotentiometric measurements of BM-$Mn_{0.25}Co_{0.75}WO_4$ recorded in a three-electrode configuration in 0.5 M H_2SO_4 at 23 ± 2 °C, 50 ± 1 °C, and 80 ± 1 °C. The applied current density was sequentially increased from 2.5 to 60 mA cm^{-2}, as indicated in the figure. (b) Corresponding Tafel plots derived from the chronopotentiometric data shown in panel (a). Data were corrected for ohmic losses using the experimentally determined uncompensated resistance (Ru) values at each temperature. Potentials are reported versus RHE and currents are normalised to the geometric surface area of the electrodes.

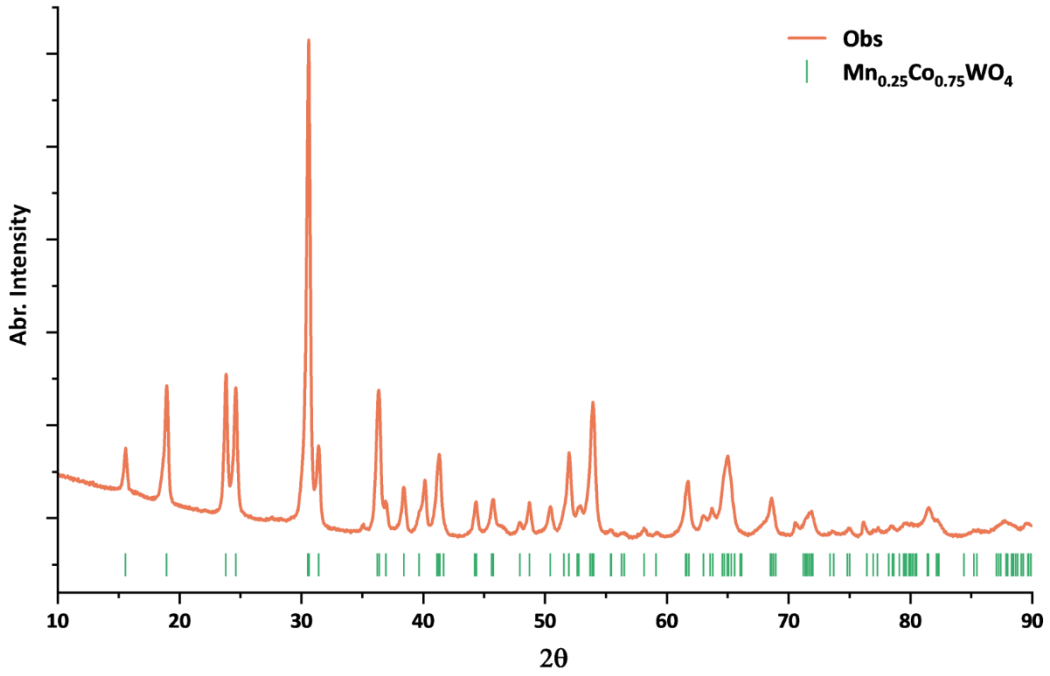


Figure S22 : X-ray diffraction (XRD) pattern of the electrode collected after the durability test.

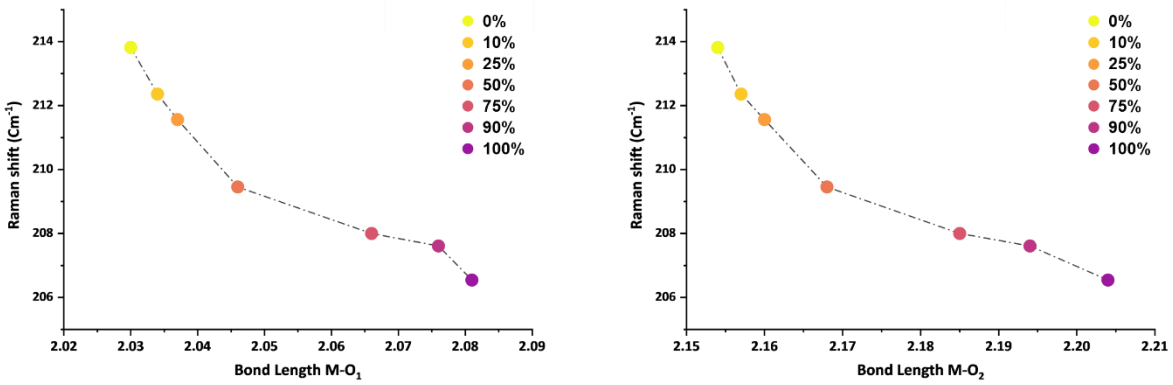


Figure S23: Linear correlation between the increasing bond distance (r) and the decreasing Raman frequency (v) for two different bonds.

(1) *Electronic properties and high-pressure behavior of wolframite-type CoWO 4 - Materials Advances (RSC Publishing) DOI:10.1039/D1MA00510C. https://pubs.rsc.org/en/content/articlehtml/2021/ma/d1ma00510c (accessed 2025-12-15).*

(2) *Temperature-dependent Raman scattering study of multiferroic MnWO4 - Hoang - 2010 - Journal of Raman Spectroscopy - Wiley Online Library. https://analyticalsciencejournals.onlinelibrary.wiley.com/doi/full/10.1002/jrs.2542 (accessed 2025-12-15).*